

COMICCRAFT

DOCUMENT 8: CODING & SOLUTION SUMMARY

FIELD	DETAILS
Repository Link / URL	https://github.com/roherror404/ComicCraft
Programming Language(s)	Python 3.12+
Framework(s) Used	FastAPI (web framework), Uvicorn (ASGI server), Jinja2 (templates), FPDF2 (PDF export), Hugging Face Diffusers (image gen), Pillow (image processing)
Key Features Implemented	■ Story prompt input with character/setting/scene/art style selection ■ 5-panel comic outline generation via Gemini 1.5 Flash ■ Rich narration and dialogue via Gemini 1.5 Pro ■ AI panel illustrations via Stable Diffusion / FLUX.1-schnell ■ Layout assembly engine combining images and text ■ Multi-page PDF export with Unicode support ■ Interactive panel-by-panel preview page ■ Mock fallback mode for API-key-free testing ■ Swagger/OpenAPI auto-documentation at /docs ■ JSON API endpoint for programmatic access
Pending / Incomplete Features	User authentication and comic history saving, Image-to-comic upload feature, Multi-language support, Social media sharing integration
Setup / Run Instructions	1. git clone https://github.com/roherror404/ComicCraft 2. python -m venv venv 3. venv\Scripts\activate (Windows) or source venv/bin/activate (Linux/Mac) 4. pip install -r requirements.txt 5. Copy .env.example to .env and add GEMINI_API_KEY=your_key_here 6. uvicorn app.main:app --reload 7. Open http://127.0.0.1:8000 in browser

Code Quality Checklist

S.NO	CRITERIA	STATUS
1	Code is modular and organized into functions/classes	■ Yes
2	Meaningful variable and function names are used	■ Yes
3	Code includes comments/documentation where necessary	■ Yes
4	Error handling is implemented for critical operations	■ Yes
5	The application runs without critical errors	■ Yes
6	Code is committed to a version control repository	■ Yes